



Monticello Archaeology Lab

Glass Lecture

Field School 2026

Th. Jefferson
MONTICELLO
ARCHAEOLOGY

Free Blown

vessels that are produced
entirely by hand, without the
use of molds or machinery



Freeblown glass tableware manufacturing, illustrations in
 Diderot's *Encyclopédie, ou dictionnaire raisonné des sciences, des arts et des métiers*, 1762-1772

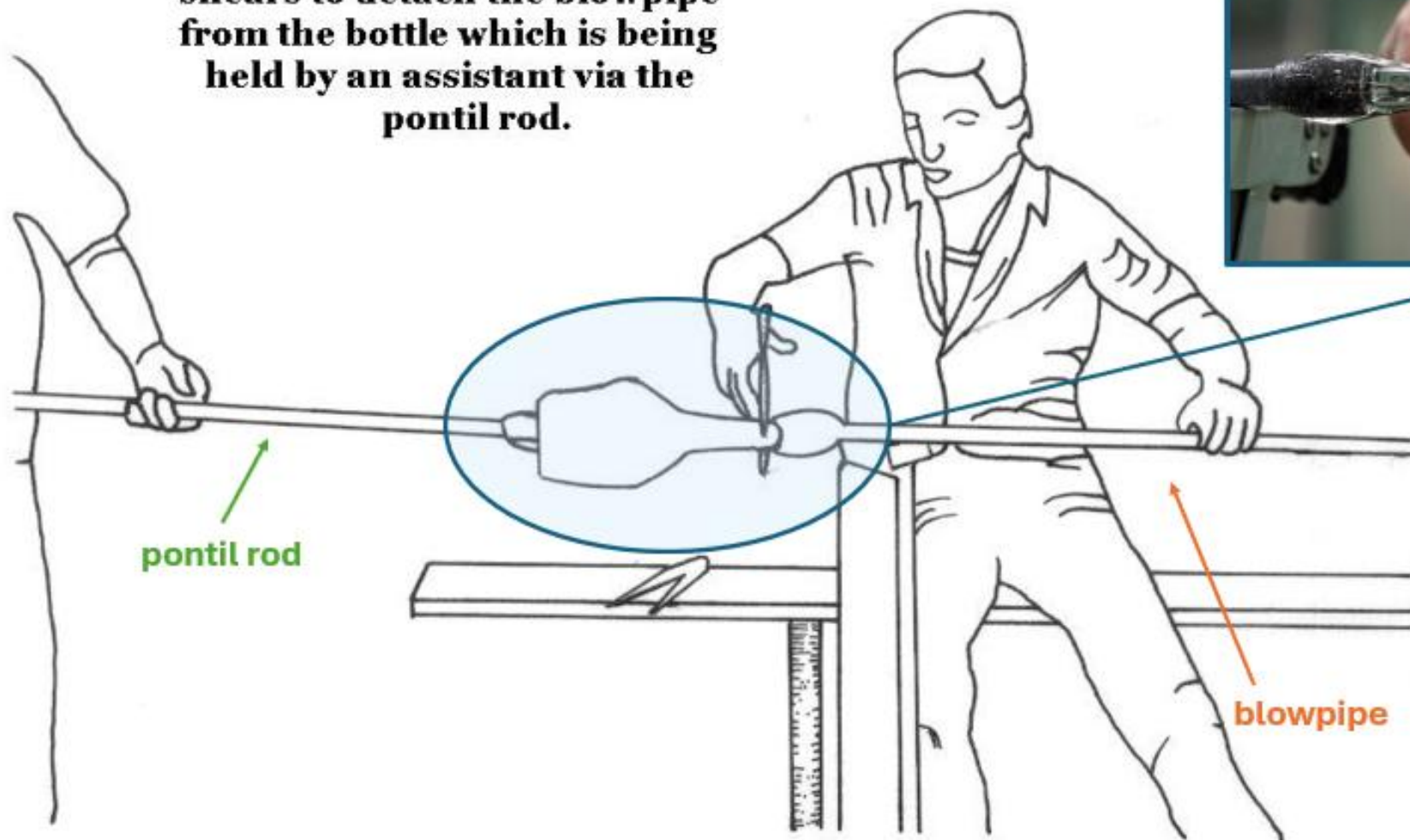
Source: <https://isleofwightglassmuseum.wikidot.com/blog:29>



Freeblown glass tableware manufacturing, illustrations in
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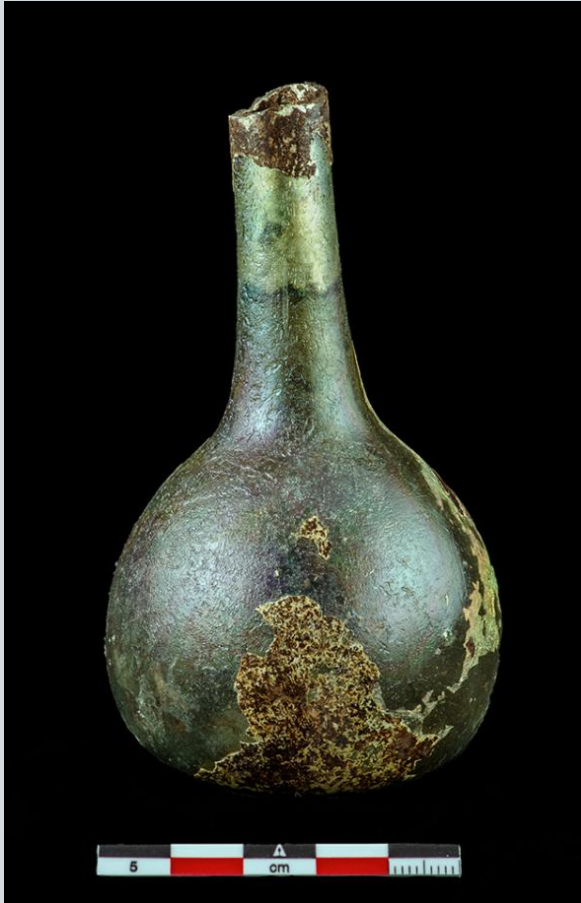
Source: <https://isleofwightglassmuseum.wikidot.com/blog:29>

Gaffer (right) at the chair using shears to detach the blowpipe from the bottle which is being held by an assistant via the pontil rod.



Model of a pontil with glass at tip

Free blown
pharmaceutical vial
(Jamestowne)



Free blown wine style
bottle
(Mount Vernon)



Free blown stemware
(Monticello)



- Organic, globular, curvilinear
 - Asymmetry
 - Pontil mark*

Pontil mark and push up on wine
style bottle from Monticello
North Yard

Mold Blown

vessels shaped by forcing molten glass into the outline of a mold through air pressure applied by mouth through a blow pipe

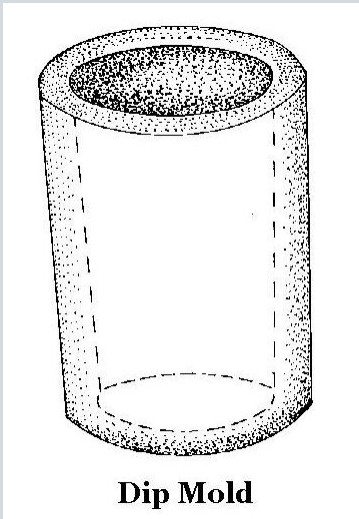


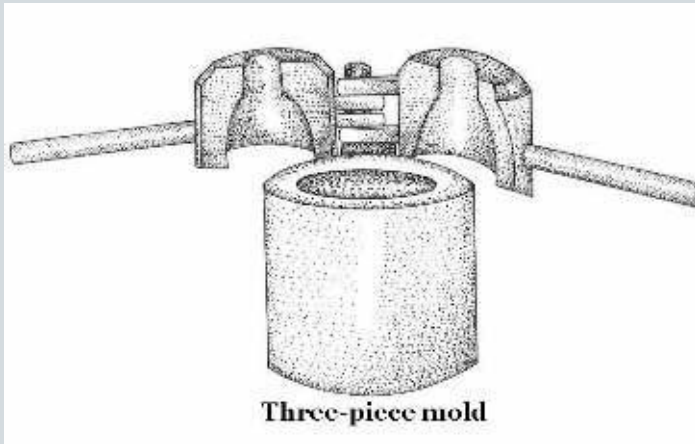
Bottles made in four-sided dip mold with hand formed shoulder, neck, and lip

Source: <https://diggingi95.com/learn-more/glassworking/mold-blown-glass/>

Trade card showing glassmakers blowing into dip molds

Source: <https://secure-sha.org/bottle/glassmaking.htm>





- Symmetry in horizontal profiles/plane
- Angles, corners, flat surfaces
- Embossing as part of vessel
 - Lettering, decoration*
- Applies or tooled finishes and their relationship to seams
- Seam lines
- Pontil scars

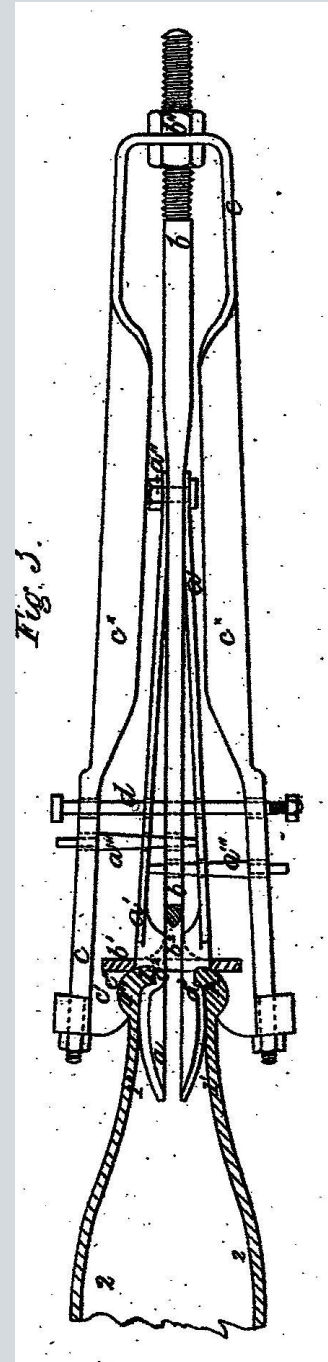
Source:
<https://bottleinfo.historicbottles.com/body.htm>

** As opposed to decorating by adding glass elements or removing glass (such as cutting or etching) on free blown vessels*





<https://collection.sciencemuseumgroup.org.uk/objects/co19247/lipping-tool-for-glass-bottles>



Lipping Tool



A photograph of two men shaking hands in an office. The man on the left is older, with short blonde hair, wearing a dark suit, light blue shirt, and a patterned tie. The man on the right is younger, with long dark hair, wearing a white short-sleeved shirt, a blue and yellow striped tie, and a blue fanny pack. They are standing in front of a desk with papers and a bookshelf. A window with blinds is visible in the background.

Mold/manufacturing
technique

decoration

Contact Mold

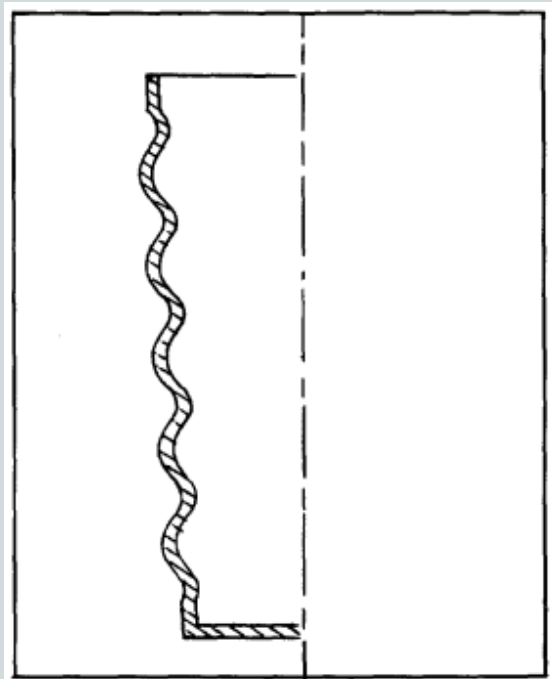


Figure 1a.
Cross-section of glass blown in contact mold.

Lorrain, 1968

Pattern Mold

Decorated part size dip mold

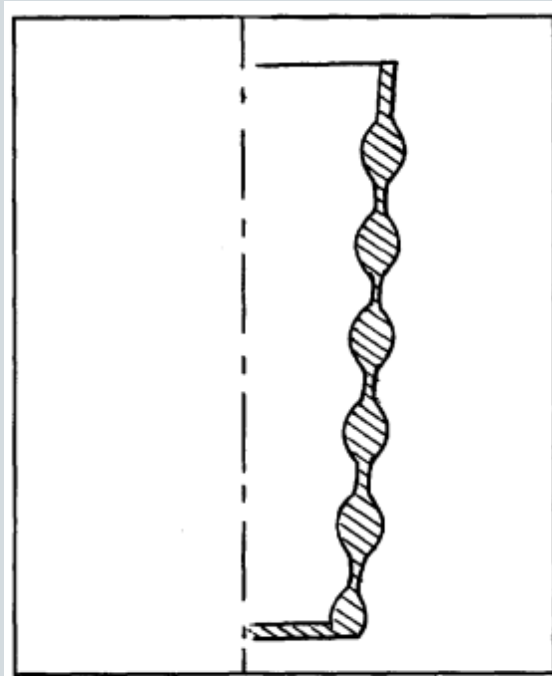
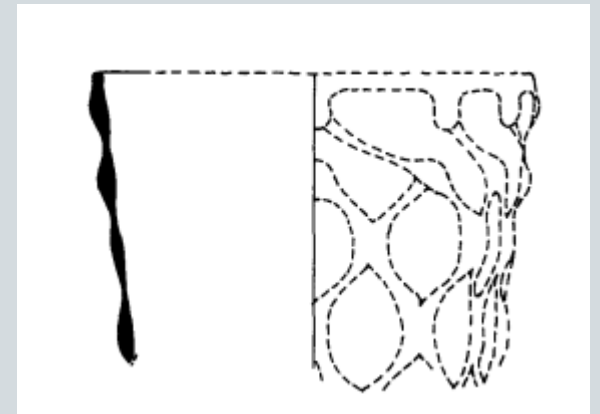


Figure 1b.
Cross-section of pattern-molded and expanded glass.

Lorrain, 1968



Jones, 1985

Optic Mold (two part process)

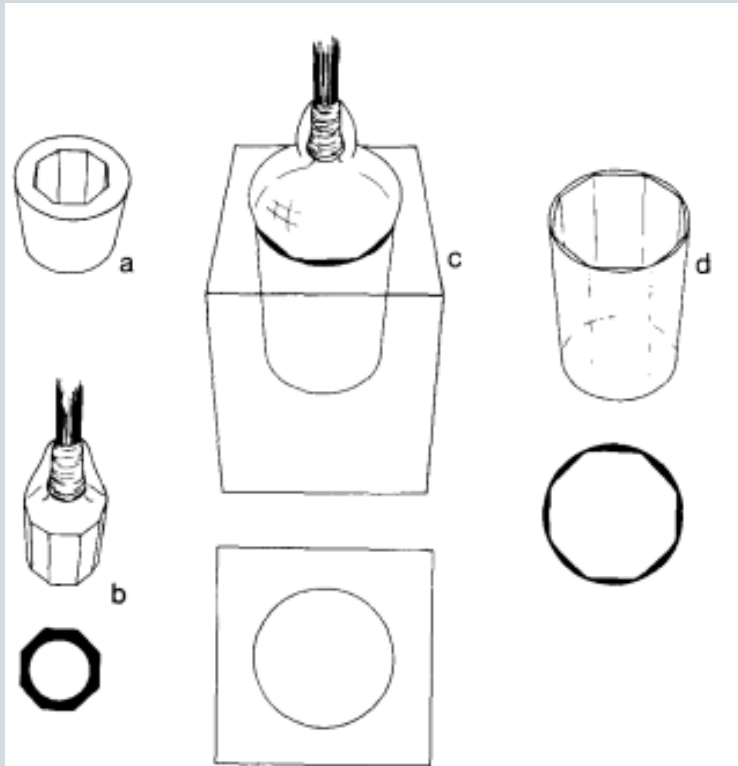
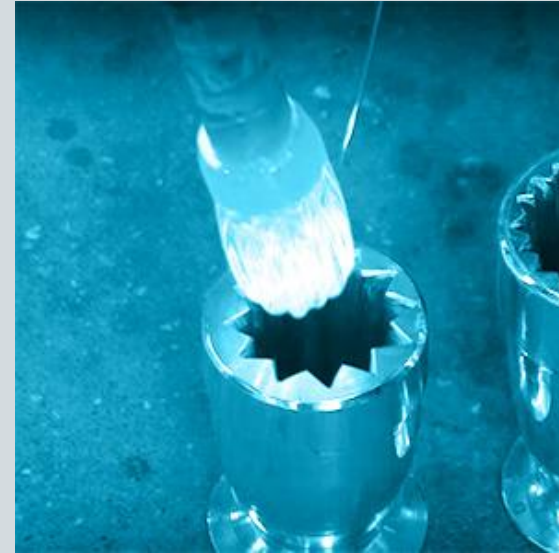


Figure 13. The process of **optic** moulding, based on Larsen, Riismøller, and Schlüter (1963: 389). (a) The part-sized mould with a pattern of panels. (b) The pattern-moulded parison. (c) The parison expanded in a smooth unpatterned dip mould. (d) Finished tumbler with the panel pattern on the inside of the object and a smooth exterior.



<https://renvenetian.cmog.org/technique/dip-optic-molding>



<https://additive-mfg.me.wisc.edu/?p=3218>

Press Mold

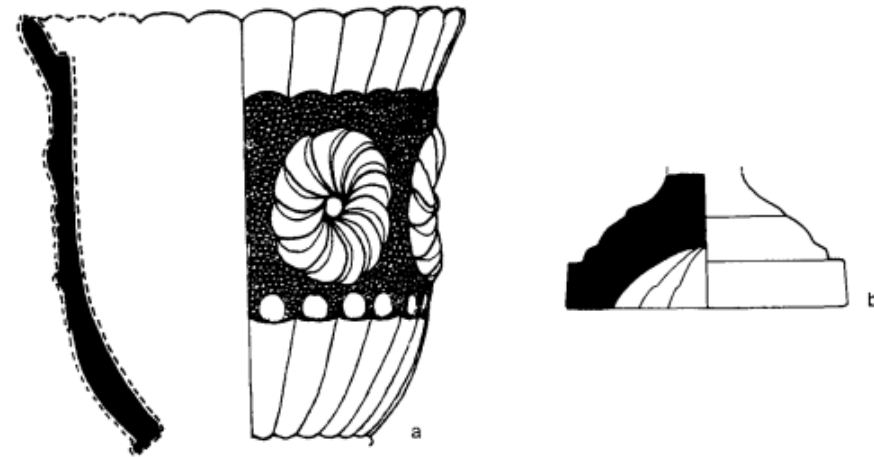
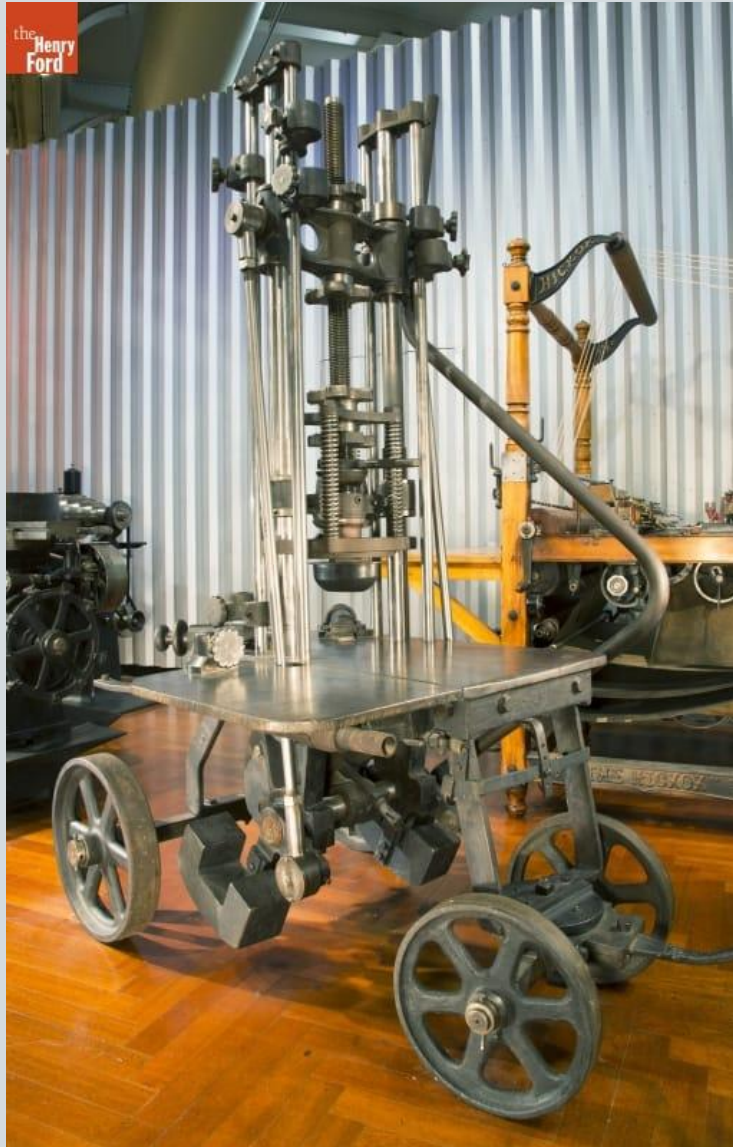


Figure 14. (a) A press-moulded bowl showing the characteristic features of press moulding. (b) "Lemon-squeezer" foot which would have been manufactured in a hand-held presser and attached to a stemmed vessel. The "plunger" was decorated with the lemon-squeezer design which is on the underside of the foot.

Mouth Blown

It is difficult to determine on the fragment level whether a vessel is completely free blown or partially mold blown with hand tooled elements.

Vessels formed by air pressure supplied by mouth

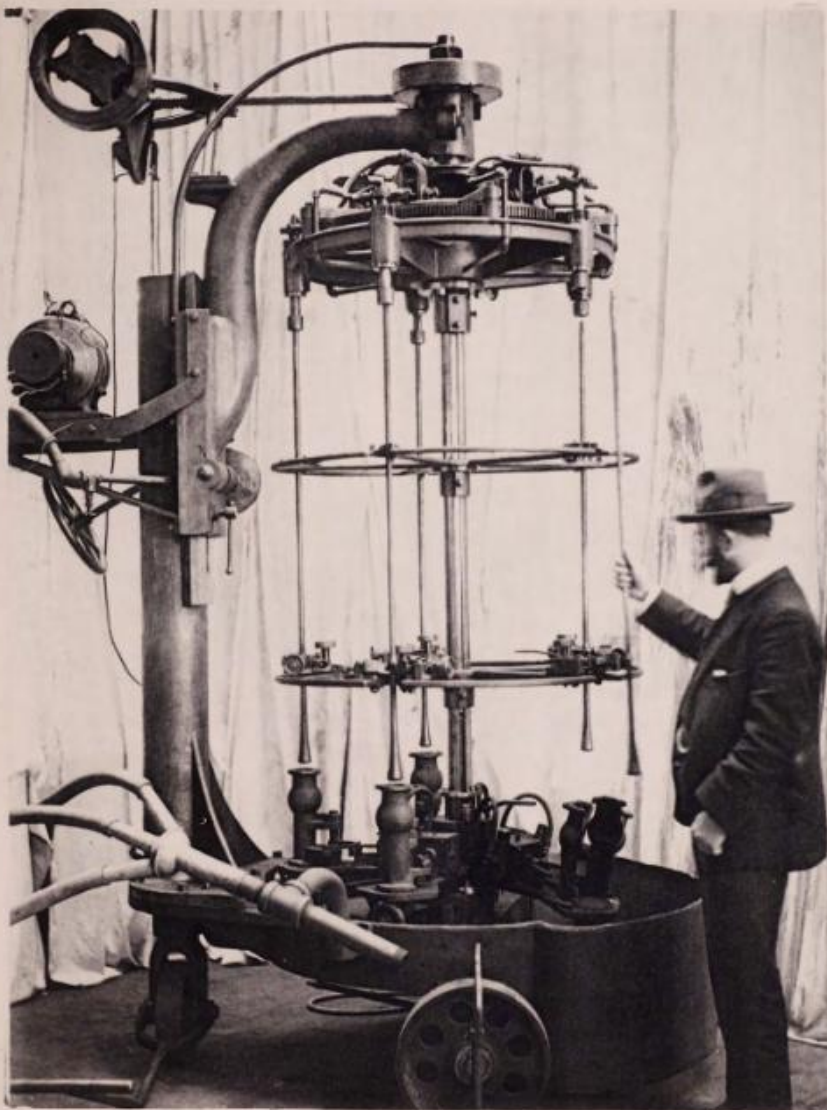
Clearly not machine made



- Historic dark green bottle glass (DGBG)/ wine-style bottles
- Colorless glass tableware

Machine Made

Vessel formed by air or pressure supplied by a machine



No. 2.—Photograph showing the Machine, with man taking the blow-iron out after the article has been blown.

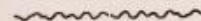
BLOWING.

The air used for blowing the glass articles is supplied by a fan or blower, conveyed to the machine by means of pipes. The machine has five spindles used for blowing the air, being admitted into the spindles by the use of an automatic cam.



FINISHING.

The method of cracking-off, annealing, cutting-off, smoothing, &c., is too well known to you for us to offer any description.



SPEED OF MACHINE.

This varies according to the size of the article being made, going slower for large articles and faster for smaller.



NUMBER OF HANDS REQUIRED.

One man to feed machine.

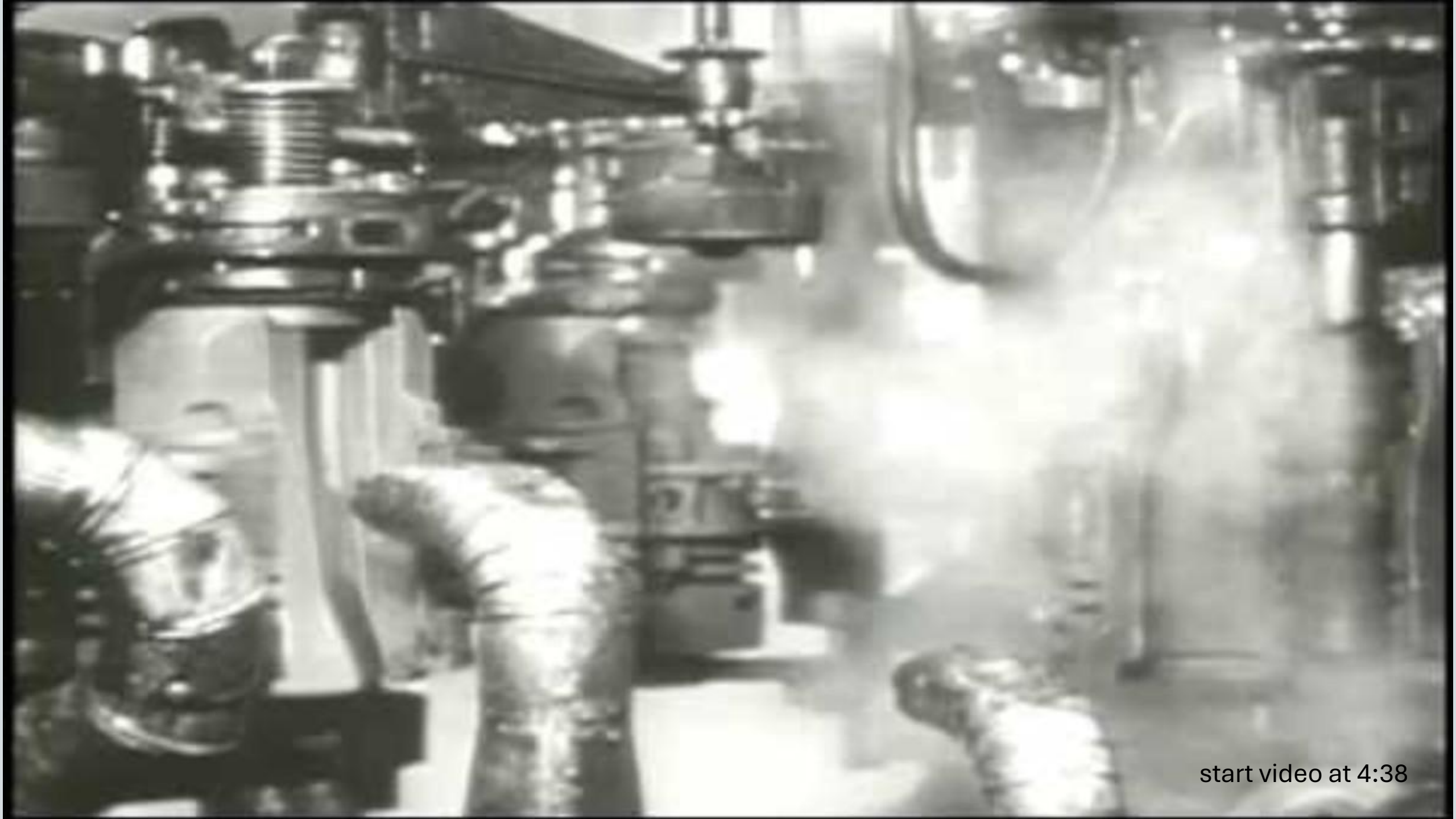
Three gatherers.

One boy to take blow-iron with article attached from the machine.



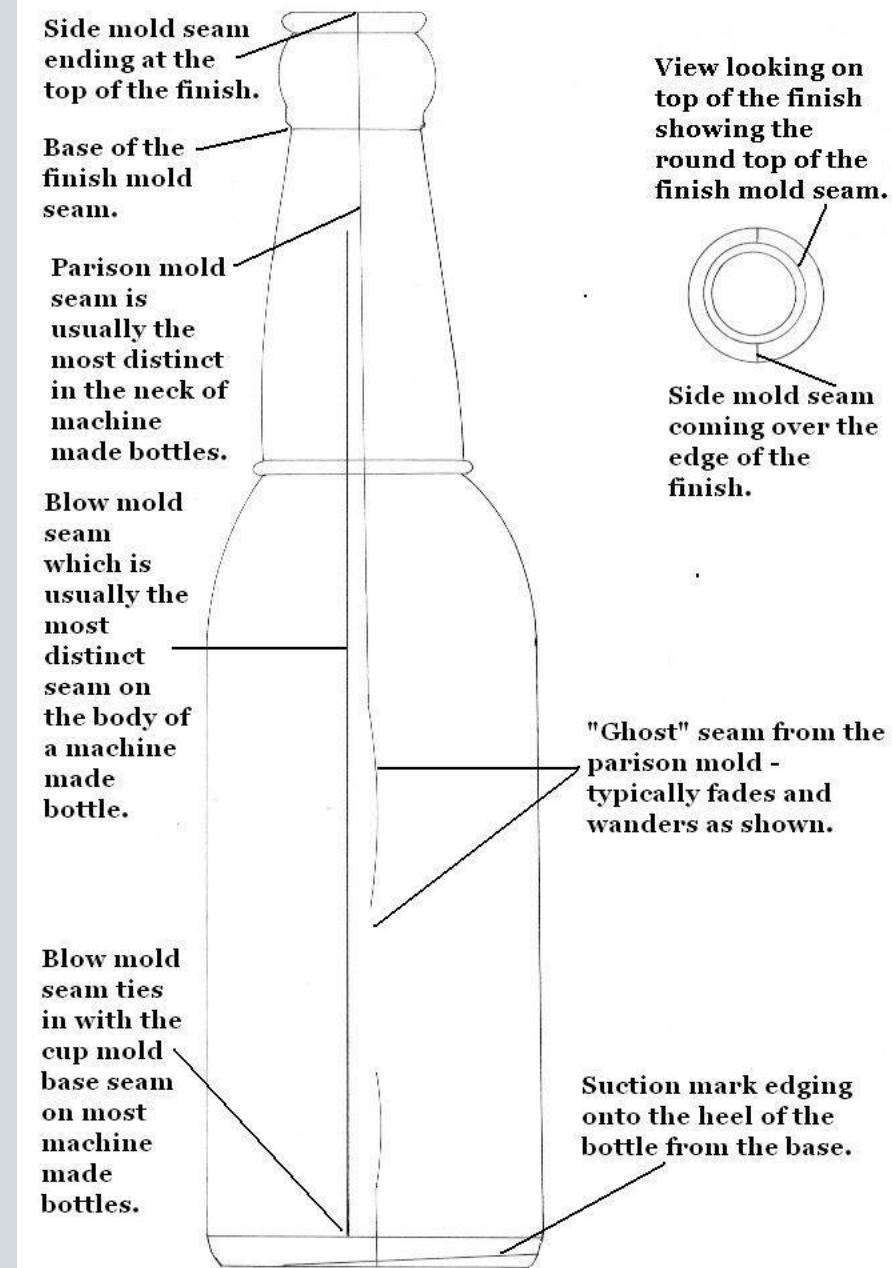
1901 Owens & Colburns' Patent Automatic Glass Blowing Machine Pamphlet

https://cmog.primo.exlibrisgroup.com/discovery/delivery/01CORNING_INST:01CORNING_I NST/1257555530004126



start video at 4:38

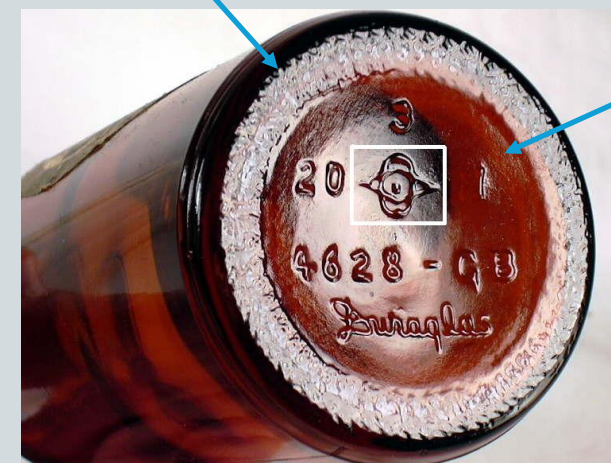
Machine Made Bottle Diagnostics



<https://sha.org/bottle/body.htm#Molded%20shape%20related%20features>



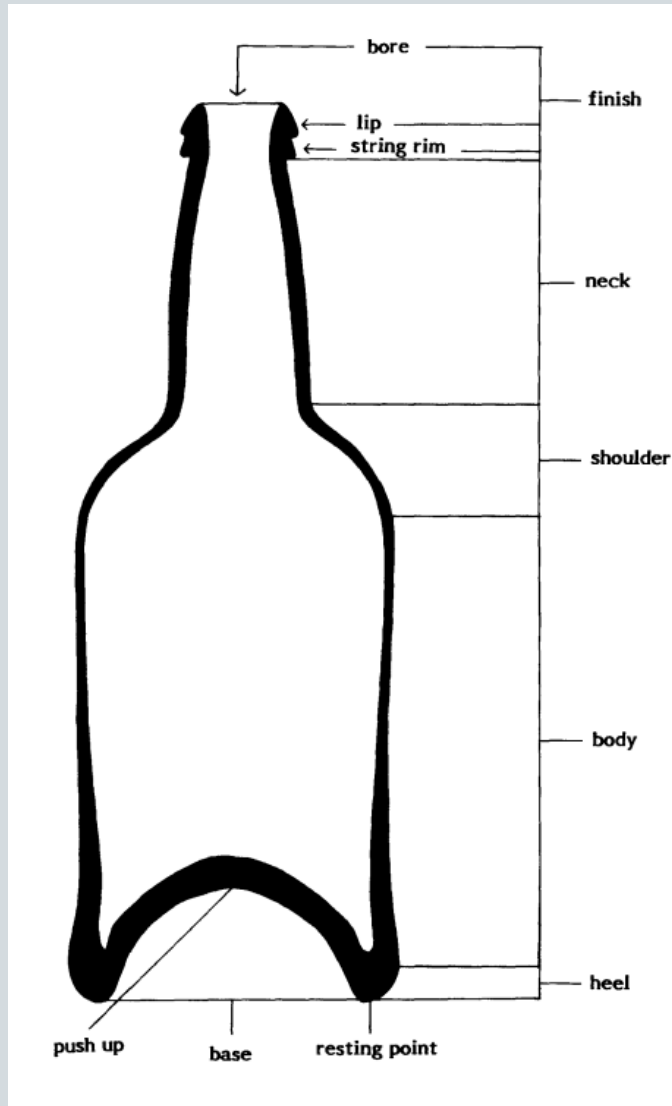
Stippling/knurling



<https://secure-sha.org/bottle/bases.htm>

Form!

Wine Style Bottle



Jones, 1985

Pharmaceutical Bottles



<https://secure-sha.org/bottle/medicinal.htm>

Tumblers



<https://diggingi95.com/archaeological-sites/historic-sites/cramp-bumm/glass-tableware-pt-2/>

Stemware

Parts of an 18th-Century Wine Glass

Bowl

This is an example of a conical bowl with a slightly curved bottom, also known as a “rounded funnel.” Other glasses of the era had bell-shaped bowls or “trumpet” bowls that flared at the top.

Merese

Glass to connect the bowl and stem

Stem with Knops

The stem is made up of bulbous knops of different widths. This one is known as an “acorn” knop, but knops of many different shapes were used on wine glass stems of the 18th century.

Base or Foot

The foot was typically wider than the bowl, and reinforced by an extra glass layer. This helped prevent chipping.

Credit: Sara Rivers Cofield of the Maryland Archaeological Conservation Laboratory.